CUTANEOUS MANIFESTATIONS OF SYSTEMIC DISEASES

Yasmin Bains, DO
Internal Medicine
Assistant Professor | Department of Clinical Sciences and Community Health

OBJECTIVES

- Identify the key cutaneous manifestations of systemic diseases that should be recognized by most physicians
- Recognize skin signs of acute, urgent and emergent diseases
- Describe the potential applications and limitations of artificial intelligence (AI) as a learning tool in dermatology
- Critically appraise an AI-generated response with an evidence-based dermatology decision support tool for a cutaneous sign of systemic disease

INTRODUCTION

- The cutaneous manifestations of systemic diseases are so numerous and varied that a single lecture could not cover them all, even in a cursory way.
- Review key cutaneous manifestations of systemic diseases that should be recognized by most physicians
- In many of the disorders presented, workup and therapy of the underlying systemic condition are essential to a favorable outcome

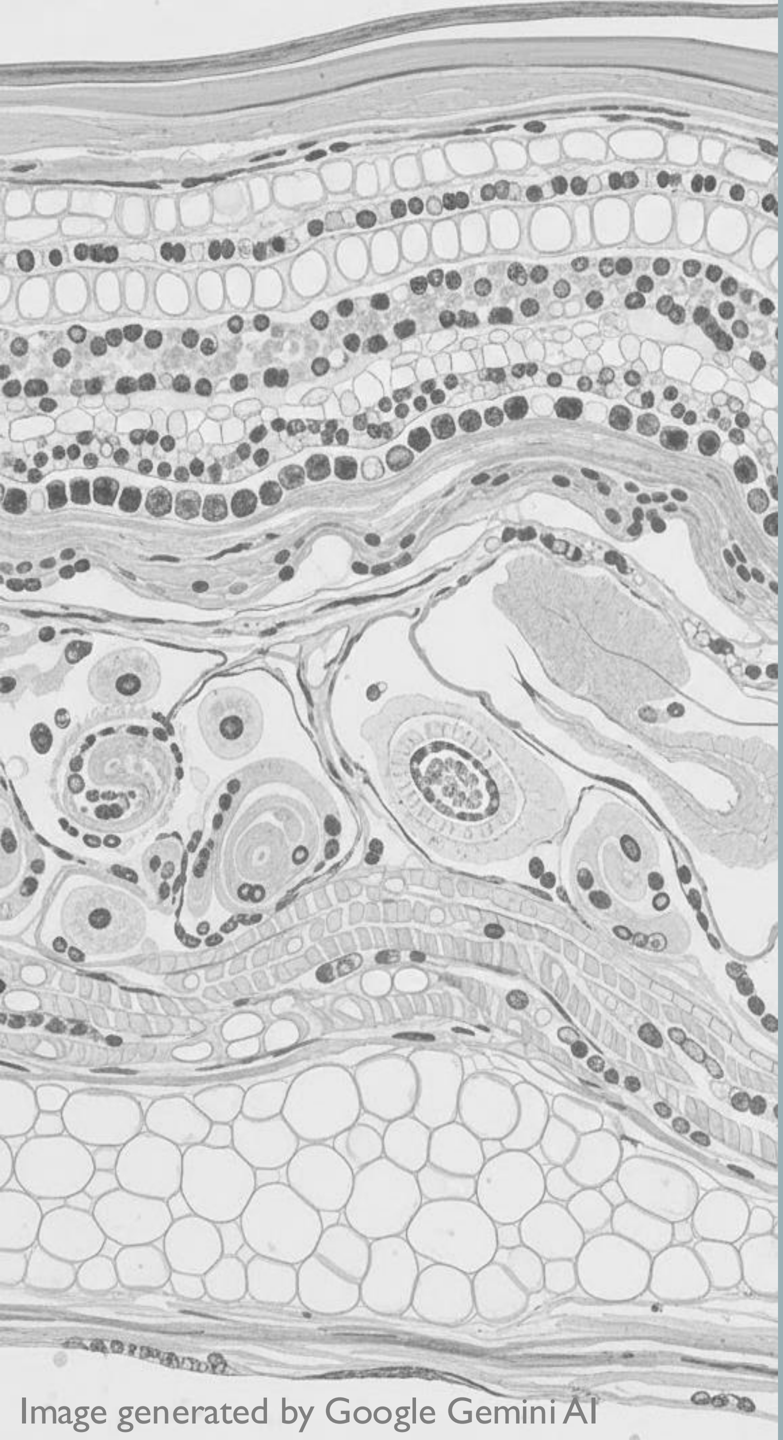
INTRODUCTION

AI in dermatology

- Advancement in deep learning models for skin cancer diagnosis, common skin conditions and chronic inflammatory skin diseases.
- With advances in AI, evaluation and adoption of AI models as they relate to bias and equity are important.
- Additional areas of exploration in clinical adoption of AI models include, ethical considerations, diagnostics and managements of complex cases, training and education for healthcare providers to effectively use AI.

SYSTEMIC DISEASES WITH CUTANEOUS MANIFESTATIONS – CATEGORIES

- **Inflammatory & Immunologic Diseases**
  - Autoimmune/Rheumatologic
  - Vasculitides
- **Endocrine & Metabolic Diseases**
- **Vitamin Deficiencies**
- **Granulomatous and Infiltrative Diseases**
- **GI and Cutaneous conditions with GI complications**
- **Systemic Infections**
- **Neurologic Diseases**
- **Renal Diseases**



INFLAMMATORY/ IMMUNOLOGIC DISORDERS

Autoimmune/Rheumatologic

Vasculitis

INFLAMMATORY/ IMMUNOLOGIC DISORDERS

Autoimmune/Rheumatologic

Dermatomyositis
Scleroderma
Cutaneous Lupus Erythematosus
Antiphospholipid Syndrome

DERMATOMYOSITIS

- Inflammatory disorder of muscle which presents with characteristic skin changes and variable systemic involvement. Cutaneous signs may precede or follow myositis
- Common Skin Manifestations
  - ADPDM - atrophic dermal papules of dermatomyositis (also known as Gottron papules)
  - Heliotrope erythema
  - Shawl sign
- Juvenile variant - characterized by calcification of skin or muscle (calcinosis cutis)

Gottron papules

Image source: VisualDx (visualdx.com)

Heliotrope erythema

DERMATOMYOSITIS

Shawl Sign: Erythematous or violaceous thin, scaly plaques are common on the upper trunk and neck

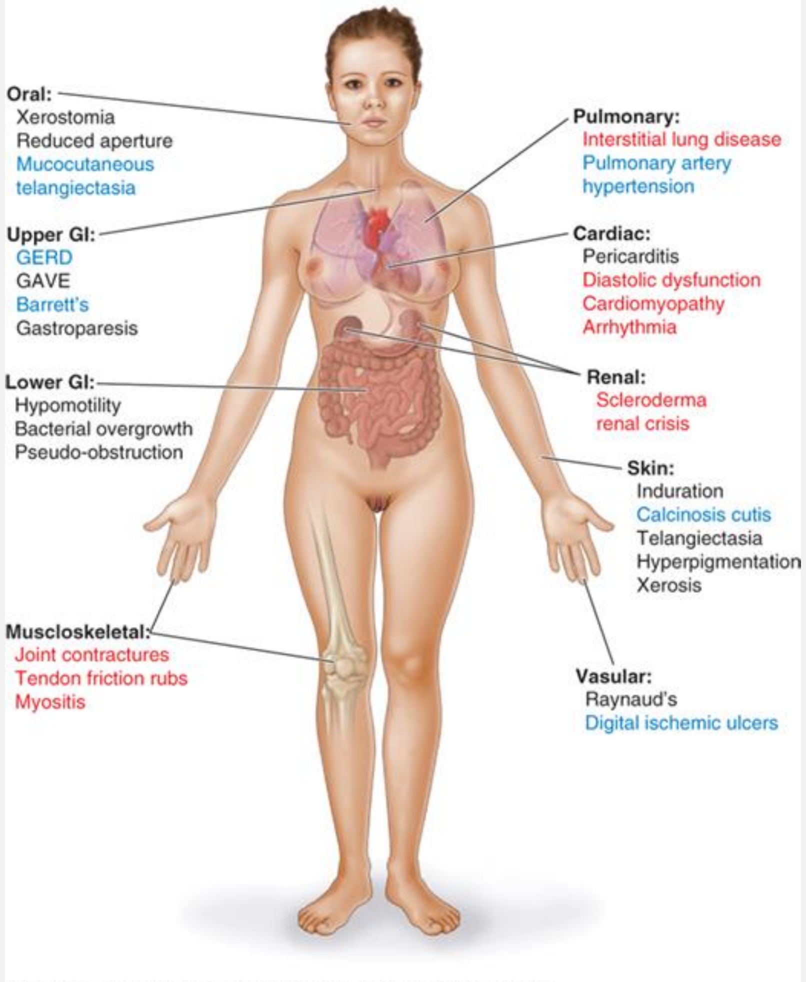
Widespread plaques in extensor forearms and periungual erythema

Calcinosis Cutis

Images source: VisualDx (visualdx.com)

SYSTEMIC SCLEROSIS (SCLERODERMA)

- Systemic Sclerosis (SSc) is a chronic, progressive connective tissue disease associated with significant morbidity and mortality.
- Multiorgan involvement: Pulmonary, cardiac, renal, skin, vascular, MSK, GI
- Distinguishing skin manifestations include thickened and indurated skin (scleroderma)
- Other skin manifestations include: sclerodactyly (skin induration and thickening distal to MCP joints), puffy fingers, digital tip ulcers, mucocutaneous telangiectasia, abnormal nailfold capillary pattern, Raynaud’s phenomenon



Source: Longo D, Fauci A, Kasper D, Hauser S, Jameson JL, Loscalzo J, Holland S, Langford C: Harrison's Principles of Internal Medicine, 22nd Edition

SYSTEMIC SCLEROSIS (SCLERODERMA)

- Diffuse cutaneous SSc (extensive skin induration) - development of interstitial lung disease (ILD) and renal crisis may develop early
- Limited cutaneous SSc – Raynaud's usually precedes sclerodactyly and other manifestations. Late-stage complications include pulmonary HTN and GI involvement
- Systemic sclerosis sine scleroderma – Raynaud's + systemic involvement without skin sclerosis
- **SSc-specific antibodies:** antinuclear, anticentromere (highly associated with LcSSc), Anti-Scl-70 (highly associated with DcSSc), Anti-RNA polymerase III (associated with scleroderma renal crisis) and others!

Image source: VisualDx (visualdx.com)

Sclerodactyly: edematous and indurated fingers. Fixed flexion contractures of proximal interphalangeal joints

Source: Longo D, Fauci A, Kasper D, Hauser S, Jameson JL, Loscalzo J, Holland S, Langford C: Harrison's Principles of Internal Medicine, 22nd Edition

Digital necrosis of fingertips

Source: Longo D, Fauci A, Kasper D, Hauser S, Jameson JL, Loscalzo J, Holland S Langford C: Harrison's Principles of Internal Medicine, 22nd Edition

Image source: VisualDx (visualdx.com)

Microstomia: decreased oral aperture

Image source: VisualDx (visualdx.com)

Sclerotic skin with associated nodules of calcinosis cutis and depigmented patches

Image source: VisualDx (visualdx.com)

Telangiectasias

LET'S SOLVE THIS

Question 1: View and answer question in Yuja Video

SYSTEMIC LUPUS ERYTHEMATOSUS

- Systemic Lupus Erythematosus (SLE) is an autoimmune condition which affects multiple organs with many clinical manifestations
- Cutaneous manifestations are common, marker for underlying systemic disease
- Cutaneous Lupus Erythematosus (CLE) – presents with multiple cutaneous phenotypes (acute, subacute, chronic) which can occur as a manifestation of SLE. Patients may have more than one phenotype

Images source: VisualDx (visualdx.com)

CUTANEOUS LUPUS ERYTHEMATOSUS

Acute Cutaneous Lupus Erythematosus (ACLE)

- Rapid onset symptoms commonly following UV exposure
- Patients with ACLE have underlying SLE
- Cutaneous presentation:
  - Malar rash – symmetric erythema/edema with fine scaling over medial cheek and bridge of nose. Spares nasolabial folds.
  - Rash can involve other areas of face, as well as neck, trunk and extremities. Poikiloderma – erythema, depigmentation may occur
  - Rare forms:
    - Toxic Epidermolytic Necrolysis (TEN)-like ACLE: in response to severe inflammatory response and causes vesiculation and skin sloughing
    - Bullous LE: antibodies against type VII collagen which leads to tense blisters

Source: Carol Soutor, Maria K. Hordinsky:
Clinical Dermatology: Diagnosis and Management of Common Disorders, 2e

Source: Carol Soutor, Maria K. Hordinsky:
Clinical Dermatology: Diagnosis and Management of Common Disorders, 2e

Bullous Lupus

CUTANEOUS LUPUS ERYTHEMATOSUS

Subacute Cutaneous Lupus Erythematosus (SCLE)

- Highly photosensitive, lesions commonly on trunk, extensor surfaces of arms
- Two variants:
  - Annular: scaly papules and plaques that coalesce with central clearing
  - Papulosquamous: Nummular erythema and scaling (can look like psoriasis)
- Associated with positive anti-Ro/SSA antibodies
- Can be triggered by medications

Annular Plaques with central clearing

Psoriasiform plaques

Images source: VisualDx (visualdx.com)

CUTANEOUS LUPUS ERYTHEMATOSUS

Chronic Cutaneous Lupus Erythematosus (CCLE)

- Most common form: Discoid Lupus Erythematous (DLE)
- Cutaneous presentation:
  - Erythematous/violaceous papules or plaques which then lead to scarring, depigmentation, hyperkeratosis due to inflammation.
  - Commonly occur on face, scalp, neck, ears (conchal bowl). May present in pattern similar to malar rash
  - Can cause scarring alopecia
  - Generalized DLE presents with discoid lesions in other areas of body too
- Positive ANA may be uncommon, work-up for SLE should be completed

Image Source: VisualDx (visualdx.com)

Image Source: VisualDx (visualdx.com)

Image Source: VisualDx (visualdx.com)

Source: Carol Soutor, Maria K. Hordinsky:
Clinical Dermatology: Diagnosis and Management of Common Disorde

Source: Carol Soutor, Maria K. Hordinsky:
Clinical Dermatology: Diagnosis and Management of Common Disorders,

NEONATAL LUPUS ERYTHEMATOSUS

- Transplacental passage of primarily anti-Ro/SSA and/or anti-La/SSB antibodies
- Mothers are often asymptomatic, but some may have SLE, Sjögren syndrome or undifferentiated connective tissue syndrome
- Annular or nummular erythematous scaly plaques commonly occurring in periorbital/forehead distribution in newborns.
- Most common cause of congenital heart block that often require cardiac pacemaker

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

ANTIPHOSPHOLIPID SYNDROME

- Acquired autoimmune thrombophilia which results in recurrent arterial or venous thrombosis, small vessel thrombosis and/or pregnancy morbidity
- Can occur in isolation or in patients with autoimmune conditions, usually SLE
- Cutaneous manifestations include:
  - Livedo reticularis - mottled reticular vascular pattern with purple discoloration of skin
  - Vasculopathy can cause painful papules or ulcers in lower extremities
- Diagnosis is clinical + lab findings

LET'S SOLVE THIS

Question 2: View and answer question in Yuja Video

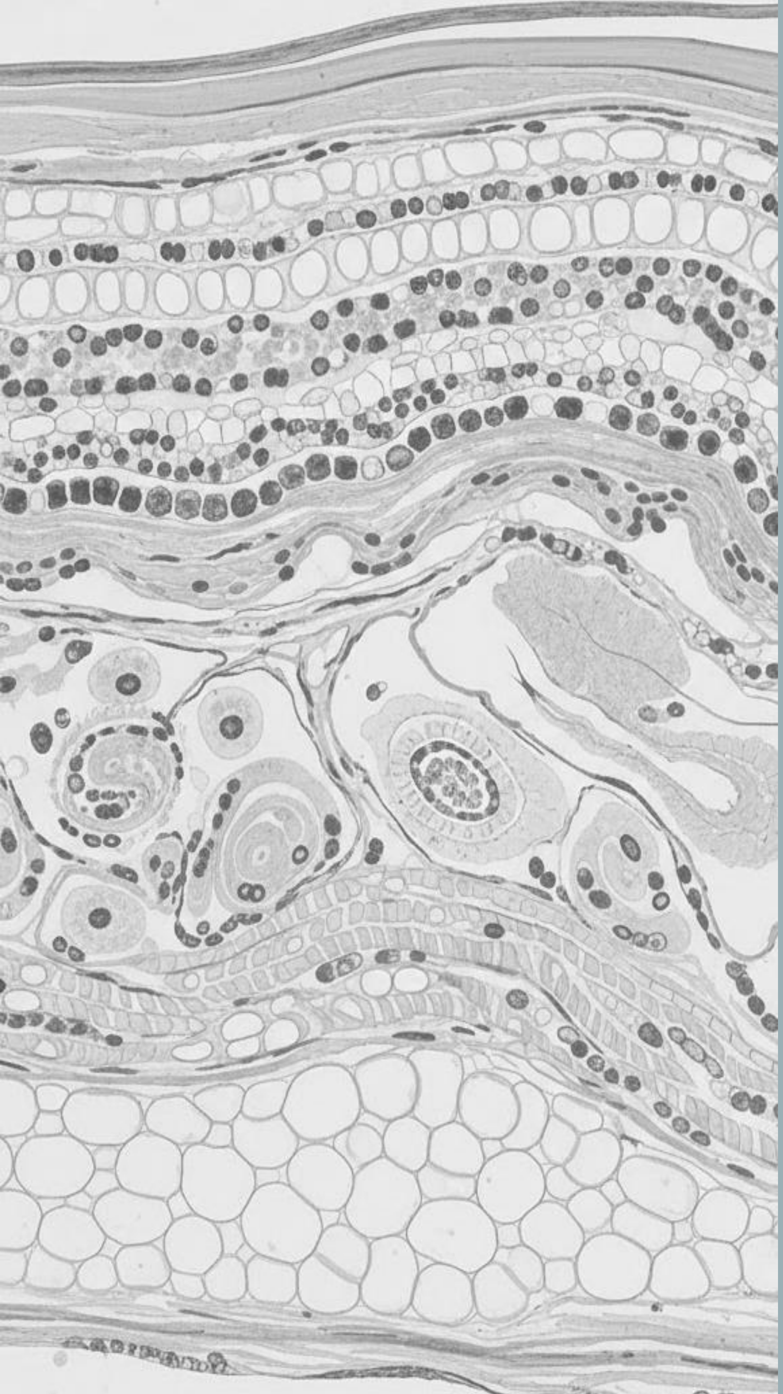


Image generated by Google Gemini AI

ARTIFICIAL INTELLIGENCE IN DERMATOLOGY

Let’s apply our knowledge of subacute cutaneous lupus erythematosus

PLEASE COMPLETE THE QUESTIONS FOR AI SECTION OF THIS LECTURE IN YUJA

ARTIFICIAL INTELLIGENCE - DEFINITIONS

- **Artificial Intelligence (AI):** “the science and engineering of making intelligent machines” – John McCarthy
- **Large Language Models (LLM):** AI which is designed to understand and generate human-like outputs based on inputs received. Built to process vast amount of datasets
  - ***Touro approved LLM models: Microsoft CoPilot, Google Gemini***
  - ***Be familiar with AI policies at your institution*** *(med school, rotations, residency/fellowship, attending)*
- **Prompt:** input provided to LLM to produce desired output
- **Prompt Engineering:** creating and refining prompts to guide the behavior of LLMs to produce high quality outputs

ARTIFICIAL INTELLIGENCE – PROMPT FRAMEWORKS

Recommended practices and elements of successful prompting:

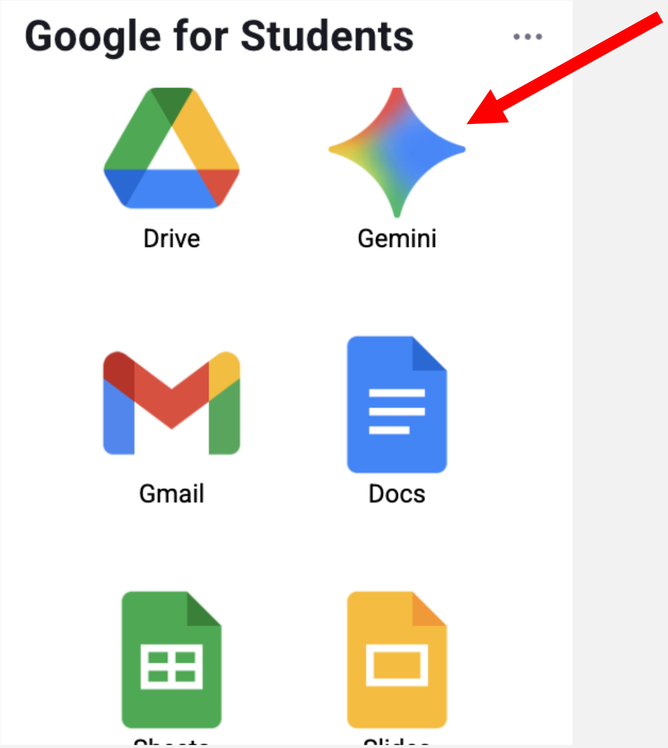
- Define clear goals
- Be specific
- Use actionable language
- Review/iterate
- Assess for bias
- Check reliability

ARTIFICIAL INTELLIGENCE – PROMPTING TECHNIQUES

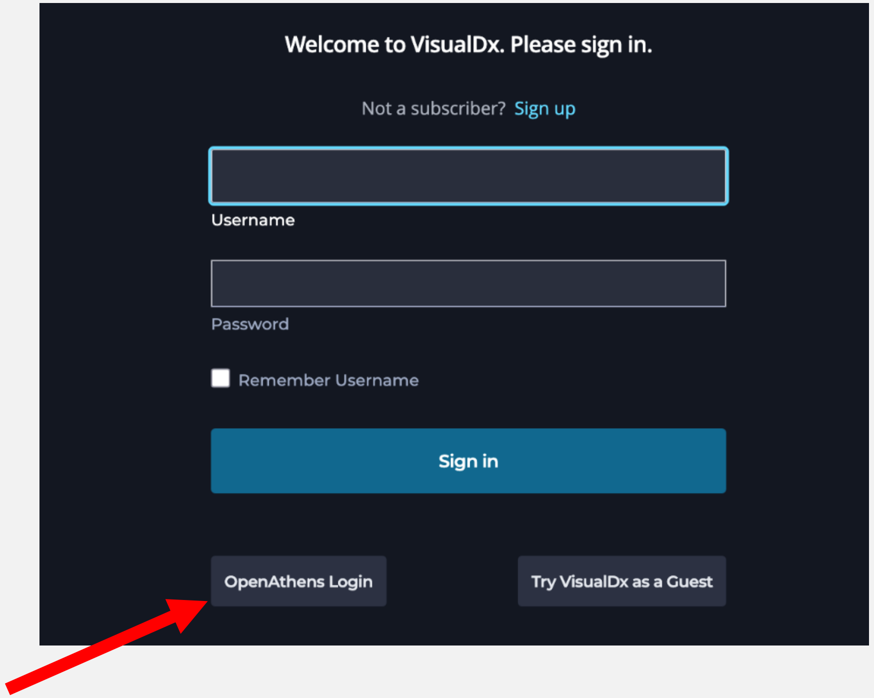
- **“Zero Shot”:** General queries/quick queries/simple facts.
  - *Example: “List the causes of pruritic rash”*
- **“One Shot” and “Few Shot”:** Provide the AI model with one (“one shot”) or more (“few shot”) examples, then ask it to solve a similar problem to teach it how to structure the answer
  - *Example: A 45-year-old female presents with pruritic rash and may have contact dermatitis. Now, list other differentials for pruritic rash*
- **“Chain of thought”:** Asking the model to provide step by step reasoning for the final output/answer. This model mimics the human method of problem solving.
  - *Example: see upcoming slides*
- **Other techniques?**

ACCESS THE FOLLOWING PLATFORMS TO COMPLETE THE NEXT ACTIVITY:

1. **Open Google Gemini – Touro Approved AI**
   Access through TouroOne Dashboard



2. **Go to https://www.VisualDx.com**
   Select OpenAthens Login
   Find Institution: “Touro University – California”
   Sign in with your TouroOne credentials

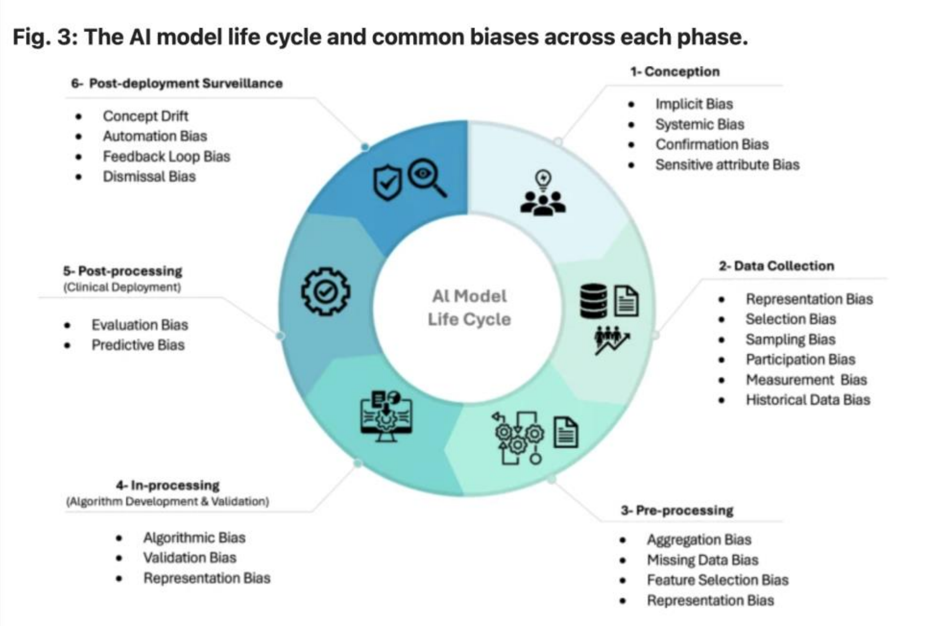


PLEASE COMPLETE THE AI ACTIVITY IN YUJA

ARTIFICIAL INTELLIGENCE – WRAP UP

AI can be a helpful tool in Dermatology, but has many limitations/challenges

- **Image representation bias:** Can lead to misdiagnosis or delay in care
- **Digital divide bias**: AI is not accessible to all populations and most major LLMs are predominantly trained using English (or languages with large datasets) which have major implications (systematic exclusion, misinformation, healthcare disparities).
- **"Hallucinations":** inaccurate/misleading outputs which lead to inaccuracies in learning and clinical decision-making
- **Oversimplification of complex/diverse clinical presentations**: this can lead to biases such as anchoring bias (i.e., not considering the diagnosis because classic findings are not met) or availability bias (i.e. not considering a diagnosis because it is less common in certain populations)



Source: Bias recognition and mitigation strategies in artificial intelligence healthcare applications. *npj Digit. Med.* **8**, 154 (2025)

ARTIFICIAL INTELLIGENCE – WRAP UP

Strategies to Mitigate Challenges with AI

- **Critical appraisal:** Evaluate outputs with your clinical reasoning skills, human judgement and validated resources
- **Evidence-based resources**: Always double check accuracy of AI generated output as well as most up to date guidelines with peer-reviewed publications and other evidence-based sources
- **Use structured prompts:** i.e. clear, step by step prompting to improve quality of outputs.
- **Iterate/modify prompts:** Develop frameworks and prompts to ensure equitable demographic representation when training AI model AND ask AI to audit its own bias
- **Stay updated:** education and trainings on how to effectively interact with AI in medicine and healthcare

INFLAMMATORY/ IMMUNOLOGIC DISORDERS

Vasculitides

IgA Vasculitis
Granulomatosis with Polyangiitis
Eosinophilic granulomatosis with polyangiitis
Kawasaki Disease
Polyarteritis Nodosa (covered in Renal Section)

HENOCH-SCHONLEIN PURPURA (IgA VASCULITIS)

- Small vessel vasculitis
- IgA immune complex, C3 and fibrin deposition in capillaries, post-capillary venules
- Commonly preceded by upper respiratory infection
- Clinical and cutaneous manifestations: palpable purpura (often on legs and buttocks), arthralgia, hematuria (IgA nephropathy), abdominal pain

Image source: VisualDx (visualdx.com)

GRANULOMATOSIS WITH POLYANGIITIS (GPA)

- Necrotizing granulomatous small vessel vasculitis which affects the upper/lower respiratory tract
- c-ANCA (PR3+)
- Palpable purpura is one of the most common skin findings
  - ulcers, subcutaneous nodules, and bullae, cutaneous ulcers
- In over 20%, skin lesions occur as the initial manifestation.
- Saddle-nose deformity, friable hyperplastic gingival tissue (“strawberry gums”), oral ulcers, ocular symptoms, renal involvement

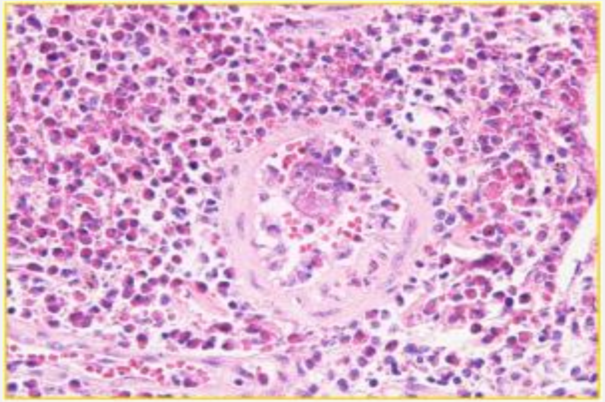
Source: Carol Soutor, Maria K. Hordinsky:
Clinical Dermatology: Diagnosis and Management of Common Disorders, 2e

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

EOSINOPHILIC GRANULOMATOSIS WITH POLYANGIITIS (EGPA)

- Necrotizing granulomatous small to medium vessel vasculitis
- p-ANCA (MPO+) in < 50% of patients
- Most commonly presents as asthma, rhinosinusitis, peripheral eosinophilia
- Palpable purpura and petechiae of the lower extremities, urticarial plaques, nodules
- Lesions show a leukocytoclastic vasculitis, infiltrative eosinophils and extravascular granulomas on skin biopsy.
- Other clinical manifestations: neuropathy, renal and cardiac (myocarditis) involvement



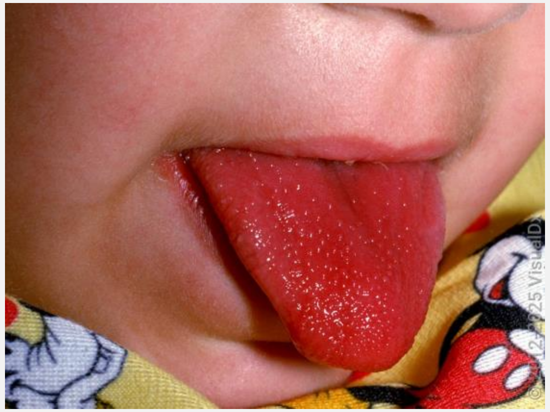
Eosinophil tissue infiltration at blood vessel wall with granuloma

Image source: Rheumatic Disease Clinics of North America, Volume 49, Issue 3, August 2023

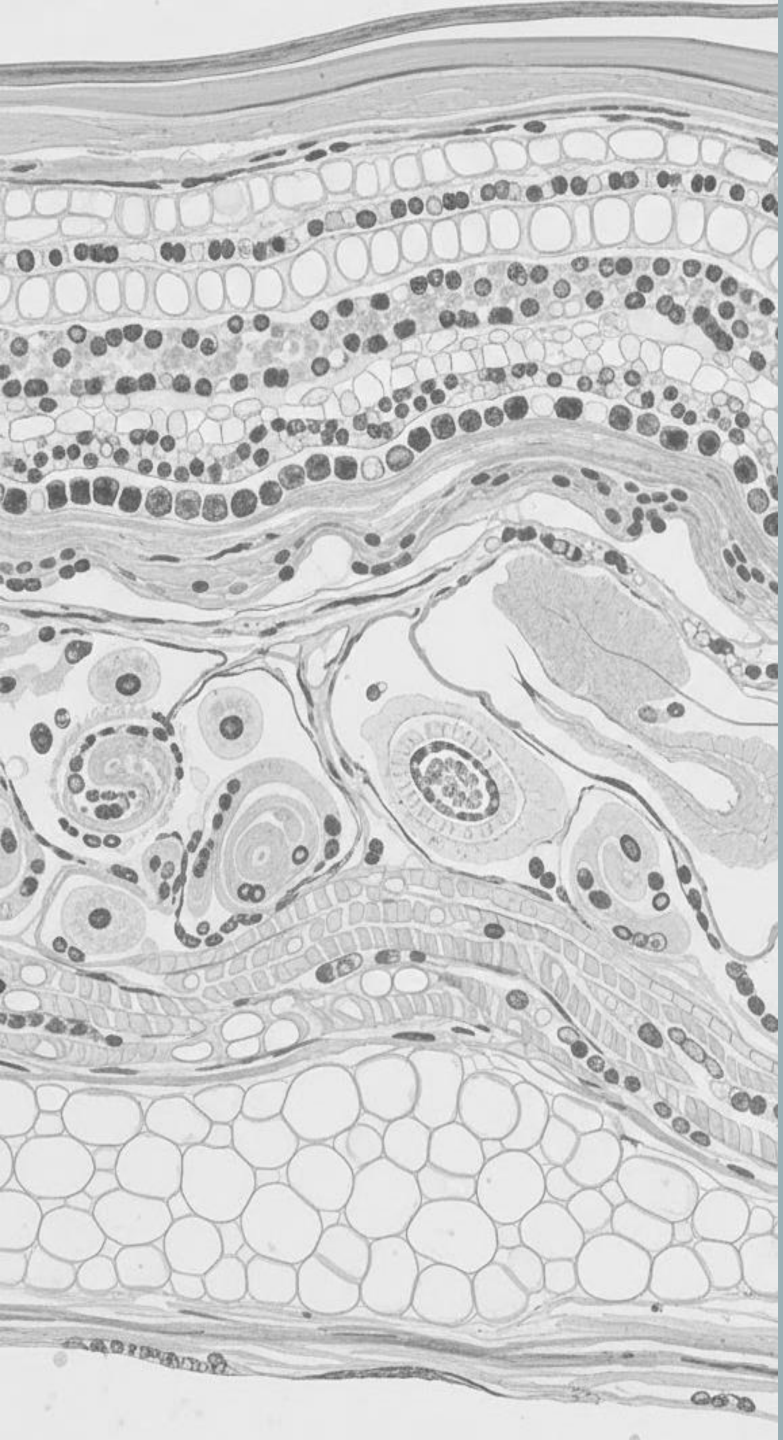
Image source: VisualDx (visualdx.com)

KAWASAKI DISEASE

- Medium vessel vasculitis, also called mucocutaneous lymph node syndrome.
- Primarily affects infants/children can be complicated by coronary artery occlusion and myocardial infarction, coronary artery aneurysms, electrocardiographic abnormalities, cardiac arrhythmias, or myocarditis.
- Diagnosis is based on clinical criteria. Symptoms include fever, bilateral conjunctivitis, cervical lymphadenopathy, perianal and scrotal erythema/scaling, oral mucosa changes ("strawberry tongue", cracked lips), exanthem, swelling/erythema of hands/feet
  - Desquamation of the palms and soles occurs in the second week of illness.
- Risk of developing coronary artery aneurysm




Images source: VisualDx (visualdx.com)



ENDOCRINE & METABOLIC DISEASES

- Diabetes
- Hyperthyroidism
- Hypothyroidism
- Xanthomas

DIABETES MELLITUS

A heterogenous group of diseases which can affect all organ systems, including the skin.

Acanthosis nigricans (AN)

- Velvety, hyperpigmented plaques on posterior/lateral neck, axillae, abdominal folds
- Commonly associated with diabetes, insulin resistance, obesity, metabolic syndrome, PCOS
- Also associated with other conditions:
  - Malignancy (gastric carcinoma)
  - Drug-induced: Niacin, steroids and others
  - Syndromic

Image source: VisualDx (visualdx.com)

Source: Carol Soutor, Maria K. Hordinsky:
Clinical Dermatology: Diagnosis and Management of Common Disorders, 2e

DIABETES MELLITUS

Necrobiosis lipoidica (NL)

- Rare, chronic granulomatous disease characterized by well-demarcated yellowish-brown or reddish-brown atrophic plaques with erythematous borders.
- Typically occurs in patients with Type 1 diabetes
- Pretibial surfaces of lower extremities are most commonly affected. Can present unilaterally or bilaterally
- Lesions can ulcerate
- Squamous cell carcinoma is a rare sequalae of chronically ulcerated lesions

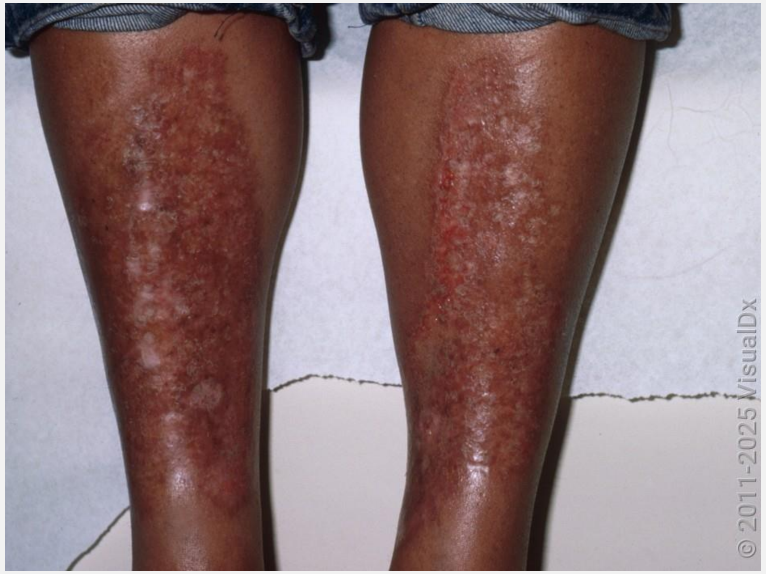


Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

DIABETES MELLITUS

Scleredema Diabeticorum

- Thickened indurated skin of the upper back, posterior neck and shoulders
- Affects patients with type 2 diabetes, more common in men
- May improve if diabetes is controlled.
- Other clinical forms of scleredema: post-infectious, paraproteinemia

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

HYPERTHYROIDISM

Graves Disease

- Circulating anti-thyroid stimulating hormone receptor antibodies
- Specific manifestations:
  - Thyroid ophthalmopathy
  - Thyroid dermopathy (pretibial myxedema)
  - Thyroid acropachy (digital clubbing, soft tissue swelling of the hands and feet, and periosteal new bone formation)

Image source: VisualDx (visualdx.com)

Images source: VisualDx (visualdx.com)

Source: Carol Soutor, Maria K. Hordinsky: Clinical Dermatology: Diagnosis and Management of Common Disorders, 2e

HYPOTHYROIDISM

- Decreased thyroid hormone due to defect in HPA axis (Hashimoto thyroiditis, thyroidectomy and many other etiologies)
- Cutaneous findings: dry/coarse skin, mottling of skin, yellowish discoloration due to carotenemia, brittle hair and nails
  - Loss of the lateral third of the eyebrows
  - Carotenemia: B-carotene metabolism is diminished (yellow appearance to skin)
- Myxedema: diffuse, non-pitting edema, "doughy" texture to the skin, and facial changes including a broad nose, macroglossia, periorbital edema, and thickened lips.

Image source: VisualDx (visualdx.com)

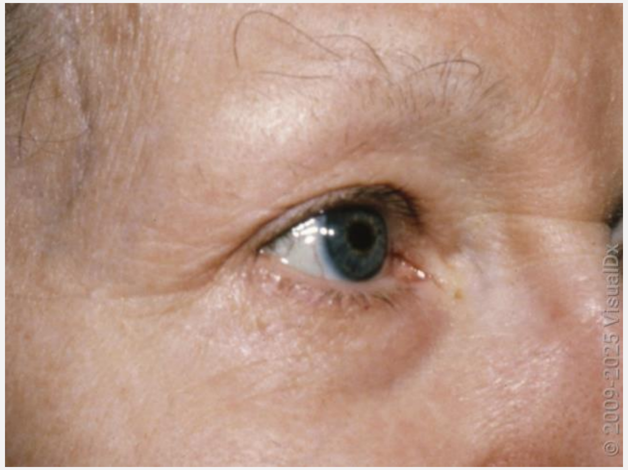


Image source: VisualDx (visualdx.com)

Image source: DermNet (https://dermnetnz.org/)

XANTHOMAS

Dermal collections of lipid laden macrophages. Can also result in extracellular lipid deposition in the dermis.

Several types of xanthomas occur with different lipid abnormalities.

Image source: VisualDx (visualdx.com)

Source: Carol Soutor, Maria K. Hordinsky:
Clinical Dermatology: Diagnosis and Management of Common Disorders

Plane (planar) xanthomas: flat yellow patches or plaques that can involve the palms, soles, neck, and chest which can be caused by familial hypercholesterolemia, hepatic cholestasis (i.e. PBC), multiple myeloma

Tuberous xanthomas: large yellow subcutaneous papules and nodules that appear on the extensor surfaces of joints, not attached to underlying tendons in the setting of hypercholesterolemia and hypertriglyceridemia

XANTHOMAS

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

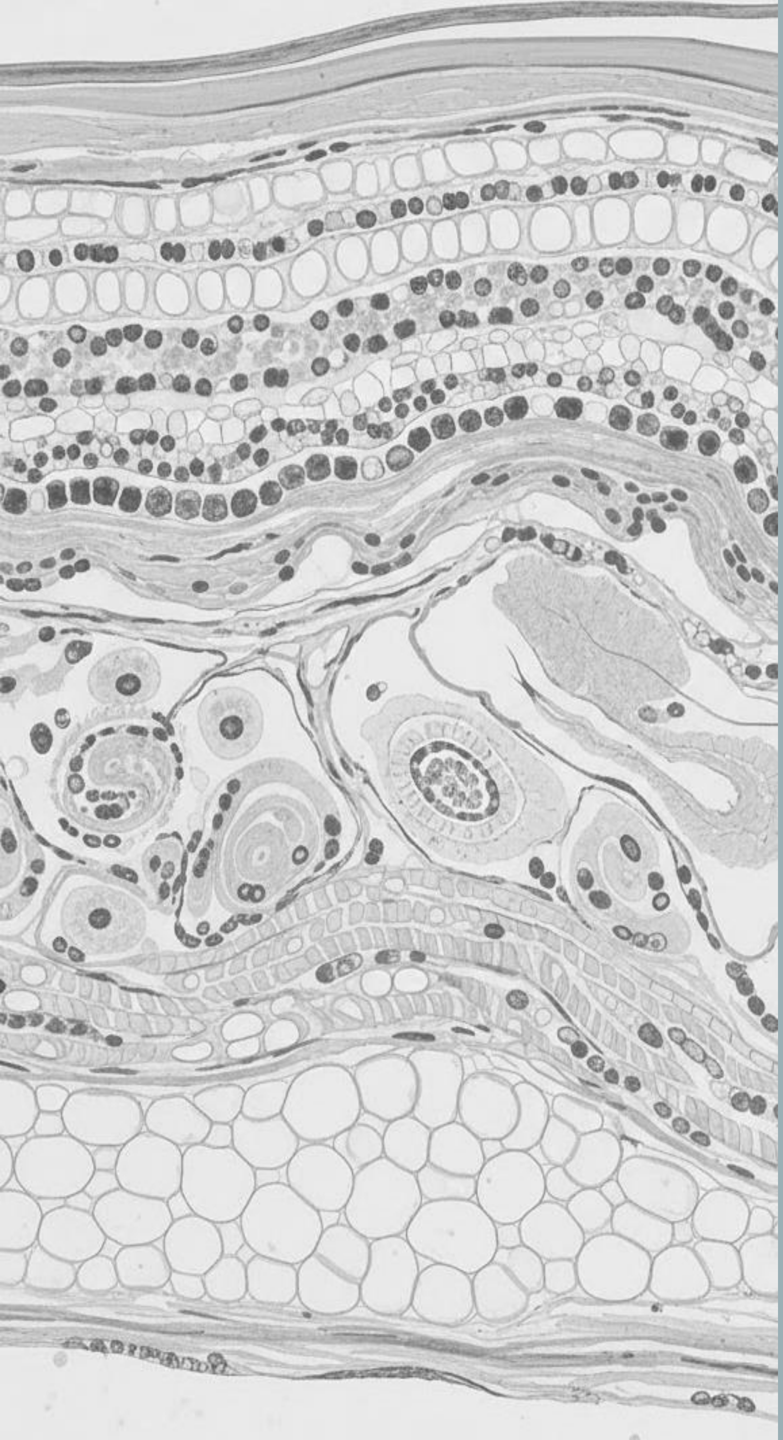
Tendinous xanthomas: Lipid deposits within extensor tendons of the elbows, ankles, knees, and hands. Most common cause is familial hypercholesterolemia resulting in high LDL levels

Eruptive xanthomas: Severe chylomicronemia and hypertriglyceridemia. Present abruptly as pruritic small yellow/orange papules over buttocks and extensor surfaces

Xanthelasma palpebrarum: papules and plaques in periocular area, commonly upper eyelid. Associated with hyperlipidemia, diabetes, thyroid disease. If lipids normal, consider monoclonal gammopathy

LET'S SOLVE THIS

Question 3: View and answer question in Yuja Video



VITAMIN DEFICIENCIES

Vitamin A
Vitamin B3
Vitamin B6
Vitamin B12
Vitamin C

VITAMIN A DEFICIENCY

- Deficiency occurs in endemic proportions in developing countries as a result of malnutrition
- In developed countries, vitamin A deficiency can occur as result of extremely calorie-restricted diet, malabsorptive diseases, cirrhosis, gastric bypass, colectomy
- Ocular presentation of vitamin A deficiency include dry eyes, Bitot's spots (white patches of keratinized epithelium on the sclera), keratomalacia (softening of the cornea)
- Cutaneous manifestations: Phrynoderma - hyperkeratotic follicular papules, usually on extensor surfaces

Image source: VisualDx (visualdx.com)

Phrynoderma

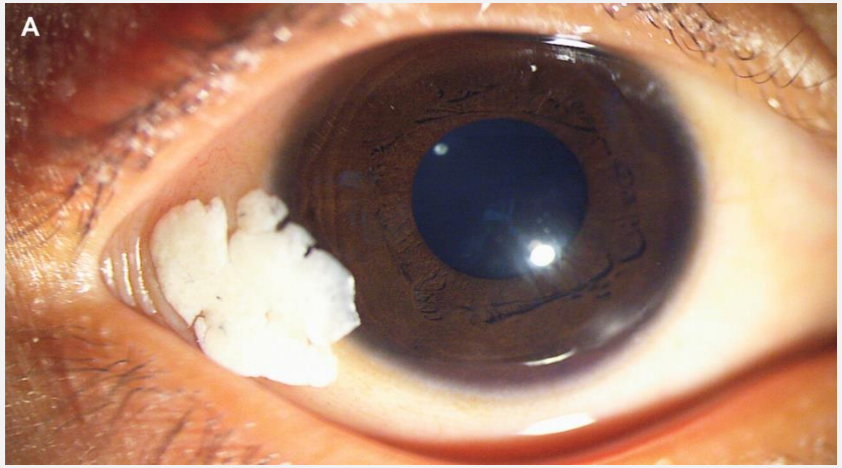


Image Source: NEJM: https://www.nejm.org/doi/abs/10.1056/NEJMicm1715354

Bitot's spots

VITAMIN B3 – NIACIN DEFICIENCY

- Niacin plays an important role in cell metabolism, namely oxidation-reduction reactions.
- Most commonly arises in areas with malnutrition and where maize is a substantial part of the diet.
- Other causes: alcohol use disorder, medications, malabsorption
- Pellagra is a disorder that results from a chronic lack of niacin
  - Four D's: diarrhea, dermatitis (photosensitive rash), dementia and death
- Which antituberculosis medication most commonly precipitates pellagra?

Hyperpigmented plaques – knees and dorsal hand

"Casal Necklace"


Crusted erythematous plaques around nose/mouth


Hyperpigmentation & scaling of face, atrophic glossitis

Images source: VisualDx (visualdx.com)

VITAMIN B6 - PYRIDOXINE DEFICIENCY

- Important cofactor in many enzymatic reactions: amino acid, lipid, glycogen metabolism as well as heme and neurotransmitter synthesis
- The classic clinical syndrome for vitamin B6 deficiency is a seborrheic dermatitis-like eruption, atrophic glossitis with ulceration, angular cheilitis (scaling on the lips and cracks at the corners of the mouth), conjunctivitis, intertrigo, sideroblastic anemia and neurologic symptoms of somnolence, confusion, depression and neuropathy.

Image Source: DermNet (https://dermnetnz.org/)

VITAMIN B12 – COBALAMIN DEFICIECY

- Causes of vitamin B12 deficiency include inadequate diet (vegan diet), lack of intrinsic factor (pernicious anemia), malabsorption syndromes (gastrectomy, chronic pancreatitis, gastric bypass, atrophic gastritis), infections (H. Pylori), meds (antacids, metformin and more)
- Cutaneous manifestations include hyperpigmented macules, atrophic glossitis, oral ulcers. Can also cause fingernail and toenail discoloration

Brown pigmentation of nails , hyperpigmentation of phalanges

Atrophic glossitis – depapillated plaques on tongue

Images source: VisualDx (visualdx.com)

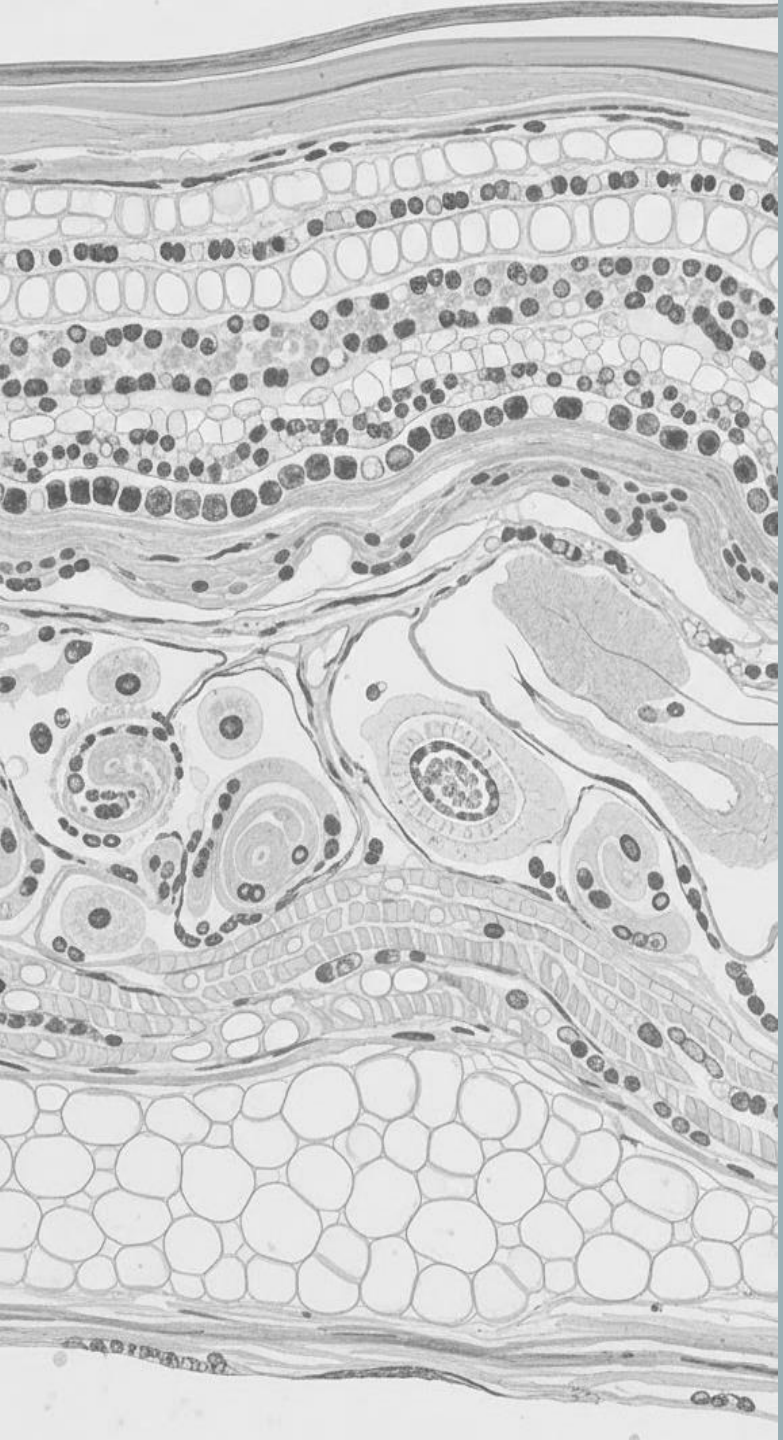
VITAMIN C DEFICIENCY

- Vitamin C is a cofactor in numerous collagen synthesis reactions, essential role in collagen and amino acid synthesis, antioxidant properties, immune function
- Dermatologic manifestations result from defects in connective tissue development
- Severe vitamin C deficiency results in scurvy, which is characterized by gingival bleeding and perifollicular petechiae with corkscrew hairs.
- Other features: purpura, petechiae, spontaneous hemorrhages into muscles, soft tissues and joints, poor wound healing, secondary infection

Edema and Hemorrhage of gums, loss of dentition

Perifollicular hemorrhage, purpura and ecchymoses

Images source: VisualDx (visualdx.com)



GRANULOMATOUS AND INFILTRATIVE DERMATOSES

Sarcoidosis

Amyloidosis

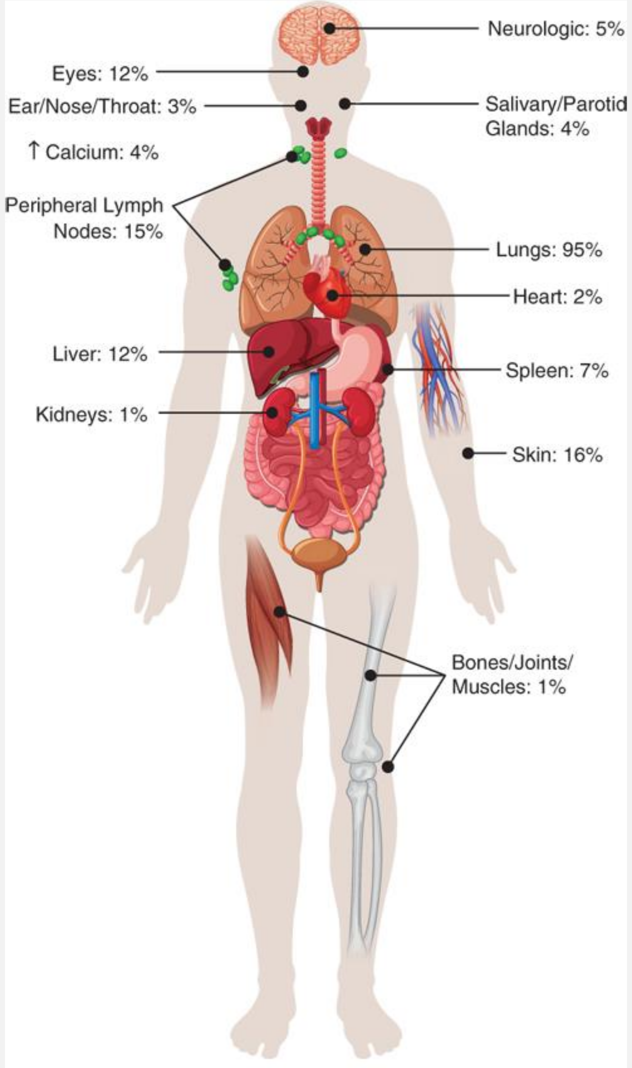
Mastocytosis

LET'S SOLVE THIS

Question 4: View and answer question in Yuja Video

SARCOIDOSIS

- Chronic inflammatory disease characterized by non-caseating granulomas
- Can affect multiple organs in the body
  - Pulmonary > 90% of cases
    Mucocutaneous >30%
- Cutaneous manifestations:
  - Macules, papules, plaques, nodules; hyperpigmented (but can also be as hypopigmented); erythematous or annular lesions on face, head, neck, upper trunk, extremities
  - Lupus Pernio – found on areas affected by cold (nose, ears, lips, face)
  - Erythema Nodosum



Source: Longo D, Fauci A, Kasper D, Hauser S, Jameson JL, Loscalzo J, Holland S, Langford C: Harrison's Principles of Internal Medicine, 22nd Edition

Image source: VisualDx (visualdx.com)

Papules and nodules on face, nose, mouth

Images source: VisualDx (visualdx.com)

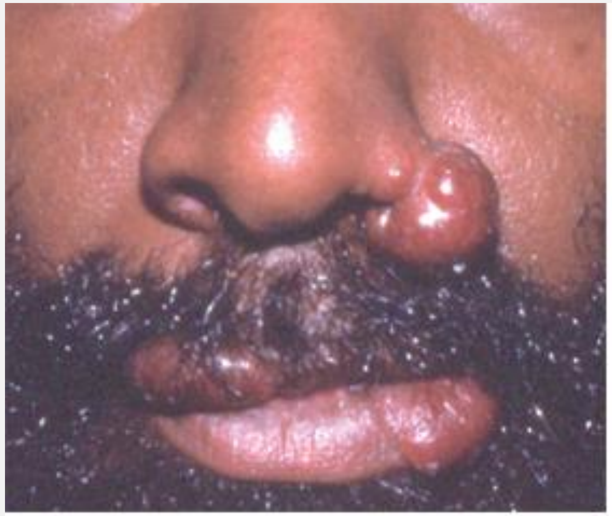
Annular plaques with hypopigmented centers

SARCOIDOSIS

Lupus Pernio: Centro-facial involvement. Associated with respiratory tract involvement

Erythema Nodosum: Tender, erythematous hyperpigmented nodules on shins

Lofgren Syndrome: fever, erythema nodosum, ankle arthralgias, bilateral hilar lymphadenopathy) can present as acute onset symptoms in up to 10% of cases



Lupus Pernio

Source: Carol Soutor, Maria K. Hordinsky:
Clinical Dermatology: Diagnosis and Management of Common Disorders, 2e

Images source: VisualDx (visualdx.com)

Erythema Nodosum

AMYLOIDOSIS

- Protein misfolding disorders characterized extracellular aggregation of misfolded protein fibrils in tissues/organs
- There are several forms of local and systemic amyloidosis
- In AL amyloidosis and amyloidosis associated with multiple myeloma (MM), amyloid fibrils consisting of immunoglobulin light chains are deposited in tissues.

B

Source: Longo D, Fauci A, Kasper D, Hauser S, Jameson JL, Loscalzo J, Holland S, Langford C: Harrison's Principles of Internal Medicine, 22nd Edition

MASTOCYTOSIS

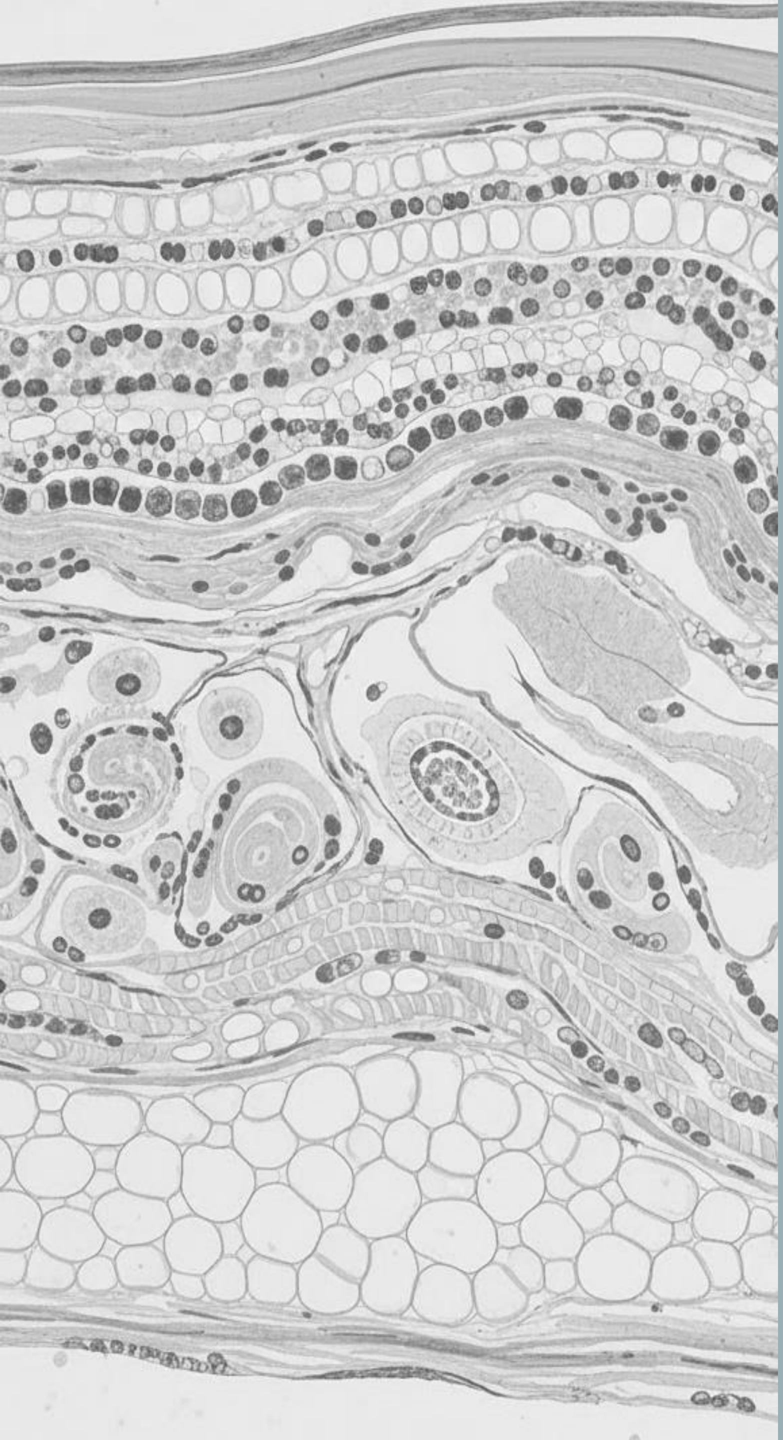
- Infiltration of clonally expanded mast cells into the skin and other organs
- Can be cutaneous with or without systemic involvement (or systemic without cutaneous disease)
- Cutaneous manifestations:
  - Urticaria pigmentosa: pink-brown macules and papules typically on trunk
  - Mastocytoma: papules, plaques, nodules (1-3 lesions).
- **Darier sign:** Thermal, mechanical or chemical stimulation of individual lesions leads to urticaria or blistering
- Other symptoms: pruritus, flushing, tachycardia, abdominal pain, diarrhea

Image source: VisualDx (visualdx.com)

Urticaria Pigmentosa

Image source: VisualDx (visualdx.com)

Mastocytoma



GASTROINTESTINAL DISEASES

Inflammatory Bowel Disease (IBD)

Celiac Disease

Peutz-Jeghers Syndrome

Carcinoid Syndrome

INFLAMMATORY BOWEL DISEASE

- Aphthous stomatitis is a nonspecific manifestation of inflammatory bowel disease (CD > UC)
  - Crohn's disease (CD): lesions may appear as noncaseating granulomas (granulomatous cheilitis)
  - Ulcerative colitis (UC) may be indistinguishable from canker sores.
- Other skin manifestations:
  - Erythema nodosum
  - Pyoderma gangrenosum

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

INFLAMMATORY BOWEL DISEASE

- **Pyoderma gangrenosum:** inflammatory, neutrophilic skin disease in patients with IBD
- The lesions are distinguishable from other ulcers by the presence of deeply violaceous, undermined borders; craterlike holes; pustules; purulent drainage
- May occur at sites of trauma.

Image source: VisualDx (visualdx.com)


Source: Carol Soutor, Maria K. Hordinsky:
Clinical Dermatology: Diagnosis and Management of Common Disorders, 2e

DERMATITIS HERPETIFORMIS

- Autoimmune immunobullous disease associated with a gluten-sensitive enteropathy (celiac disease)
- IgA immune complex deposition in the papillary dermis
- Skin lesions begin as papulovesiclar eruption that are very pruritic and quickly broken by scratching, leaving only excoriations and crusts

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

LET'S SOLVE THIS

Question 5: View and answer question in Yuja Video

PEUTZ-JEGHERS SYNDROME

- Rare, autosomal dominant syndrome
- Oral and/or perioral hyperpigmented lesions and intestinal hamartomatous polyps (benign)
- Pigmented macules can develop on the palms, fingers, soles, and toes and in areas around the mouth, nose, and rectum

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

PORPHYRIAS

Porphyria Cutanea Tarda (PCT)

- Porphyrias are hereditary disorders - defects in heme synthesis
- PCT is the most common – autosomal dominant or acquired. Characterized by photosensitivity, vesicle formation (especially on the dorsa of the hands), and hypertrichosis.
- Acquired forms are associated with hepatitis C, alcohol and tobacco use, elevated estrogen, iron overload, HIV

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

CARCINOID SYNDROME

- Associated with neuroendocrine tumors (commonly with tumors in the small intestine, colon, appendix)
- Can develop after hepatic metastatasis or retroperitoneal lesions → serotonin and other vasoactive substances into circulation → causes flushing and other cutaneous manifestations
  - Less common cutaneous manifestations include scleroderma-like skin changes and pigmentation changes similar to pellagra.
- Lungs are second most common site of involvement - episodic flushing associated with abdominal pain, diarrhea and wheezing

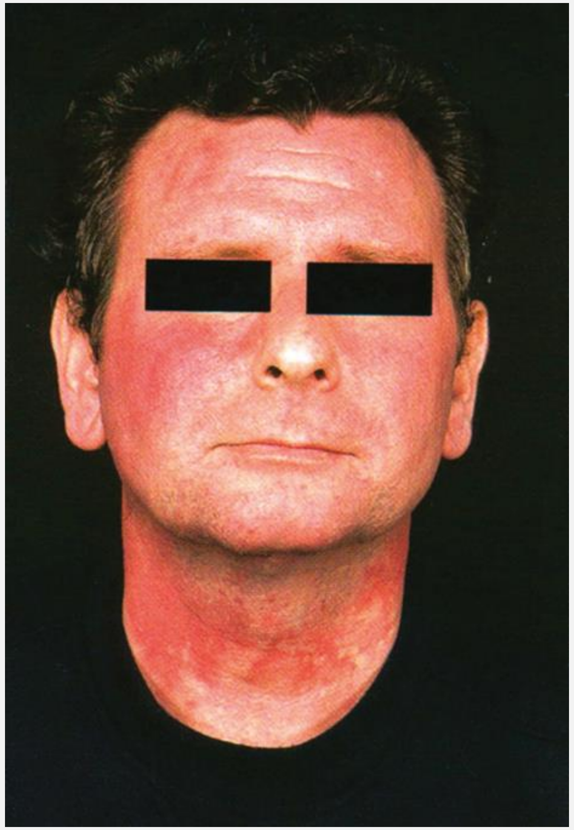
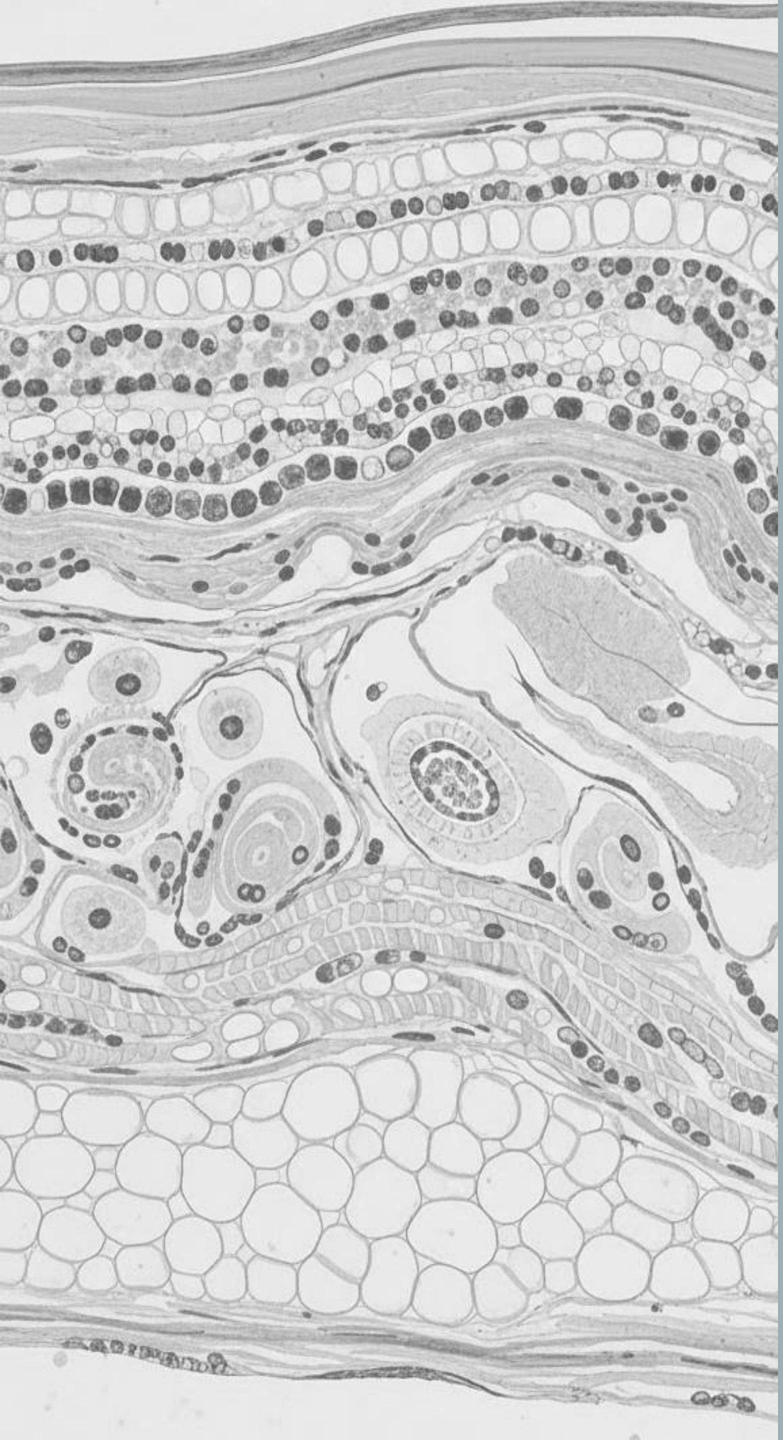


Image source: Rev Endocr Metab Disord (2016) 17:373–380. DOI 10.1007/s11154-016-9394-8



SYSTEMIC INFECTIONS

- HIV/AIDS
- Infective endocarditis
- Staphylococcal Toxic Shock Syndrome
- Staphylococcal Scalded Skin Syndrome
- Necrotizing Fasciitis
- Meningococcemia
- Rocky Mountain Spotted Fever
- Scarlet Fever
- Vibrio Infection
- Lyme Disease
- Rheumatic Fever

CUTANEOUS MANIFESTATIONS OF INFECTIONS AND OTHER DISEASES ASSOCIATED WITH ACQUIRED IMMUNODEFICIENCY SYNDROME (AIDS)

- **Opportunistic infections**
  - Viral
  - Fungal
  - Bacterial
  - Scabies
- **Kaposi Sarcoma**
- **Oral Hairy Leukoplakia**

AIDS - VIRAL

Molluscum contagiosum

- Infection of skin and mucous membranes caused by pox virus group
- In adults, more common in patients with HIV/AIDS
- Small umbilicated papules on the skin. Can occur on genital mucous membranes 2–7 weeks after contact with an infected individual.
- Initially asymptomatic, can develop pruritus and inflammation. May be larger and have extensive distribution in immunocompromised

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

Mpox

- Zoonotic infection caused by pox virus group (Orthopoxvirus)
- Endemic in Central and West Africa, but has spread to > 100 non-endemic countries since global outbreak in 2022
- Papular, vesicular, then pustular eruption on the face, trunk, and extremities.

D

Source: Longo D, Fauci A, Kasper D, Hauser S, Jameson JL, Loscalzo J, Ho
Langford C: Harrison's Principles of Internal Medicine, 22nd Edition

F

Source: Longo D, Fauci A, Kasper D, Hauser S, Jameson JL, Loscalzo J, Holland S,
Langford C: Harrison's Principles of Internal Medicine, 22nd Edition

Circumscribed lesions with central umbilication. Right image shows severe manifestations in HIV/AIDS

AIDS - VIRAL

Herpes simplex virus

- DNA viruses which affect oral cavity, lips, perioral skin (HSV-1) or genital region (HSV-2)
- Can present similarly as in immunocompetent but can also present as ulcerative erosions and widespread dissemination to other
- Acyclovir-resistant strains of herpes simplex virus have been reported in some patients with AIDS

Image source: VisualDx (visualdx.com)

Erythema Multiforme

Image source: VisualDx (visualdx.com)

Large erosive lesions intergluteal cleft

Herpes zoster virus

- infections are a common sign of HIV infection.
- In the non–HIV-infected host, herpes zoster is characterized by grouped vesicles in a dermatomal distribution. The eruption is self-limited, resolving within 1 to 2 weeks.
- In contrast, herpes zoster infection can develop into a disseminated vesicular eruption in immunocompromised patients and is a sign of viremia

Source: Carol Soutor, Maria K. Hordinsky: Clinical Dermatology: Diagnosis and Management of Common Disorde

Image source: VisualDx (visualdx.com)

Crusted plaque on upper breast (T2 dermatome) with surrounding vesicles and papules on the chest

LET'S SOLVE THIS

Question 6: View and answer question in Yuja Video

AIDS –BACTERIAL

Bacillary angiomatosis

- Vascular skin lesion caused by Bartonella henselae (cat exposure) or Bartonella quintana (body louse)
- Purple or red papules and nodules that can be mistaken for Kaposi sarcoma, cherry angioma, pyogenic granuloma
- Systemic signs and symptoms include fever, night sweats and multiple organ involvement (liver, spleen, lytic bone lesions, lung, CNS, cardiac GI)

Images source: VisualDx (visualdx.com)

AIDS – FUNGAL

Many fungal pathogens are opportunistic infections in immunosuppressed

Monilial infections include oral thrush and candidiasis of the groin.

Cryptococcosis:

- Cryptococcus neoformans - encapsulated yeast
- Infection can remain latent in the lung/hilar nodes and proliferate to cause active infection causing hematogenous spread to other organs including skin
- Diverse lesions: ulcers, nodules, papules, pustules, centrally umbilicated vesicular lesions

Endemic Mycosis: Histoplasmosis, blastomycosis, coccidioidomycosis, sporotrichosis

Cryptococcosis lesions:
Top left: umbilicated papules
Top Right: Bleeding ulcer surrounded by scaling
Bottom: Vegetating nodule

Images source: VisualDx (visualdx.com)

AIDS – KAPOSI SARCOMA (KS)

- Slowly progressive vascular neoplasm associated with human herpesvirus 8 (HHV-8)
- Four epidemiologic forms:
  - Classic KS: older men of commonly Mediterranean or eastern European Jewish background; with no precipitating factors
  - African Endemic KS: all ages; no precipitating factors
  - Iatrogenic KS: Organ transplant
  - AIDS-associated KS
- Papules, patch which can develop into plaques or nodules which coalesce or ulcerate.
- In AIDS-associated KS, lesions can occur in any surface of the body, including mucocutaneous and visceral involvement.

Images source: VisualDx (visualdx.com)

AIDS – ORAL HAIRY LEUKOPLAKIA

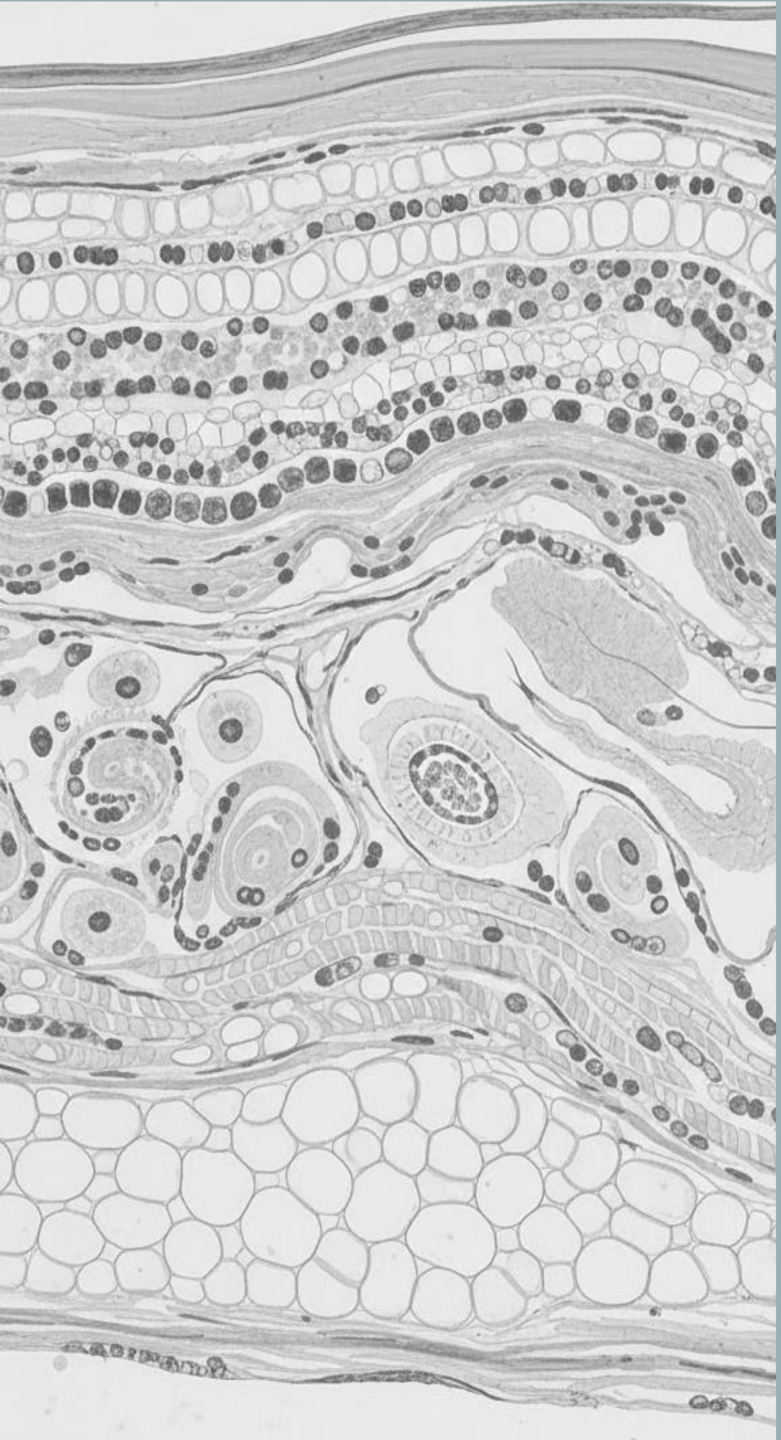
- Painless, white patches of the oral mucosa, usually lateral tongue
- Note: oral leukoplakia is a potential malignant condition versus oral hairy leukoplakia is not premalignant and associated with Epstein- Barr Virus in HIV patients
- Oral hairy leukoplakia can be distinguished from thrush (oral candidiasis) in that the lesions cannot be rubbed off, as they can be in thrush.

Image source: VisualDx (visualdx.com)

Oral hairy leukoplakia

Image source: VisualDx (visualdx.com)

Oral candidiasis (thrush)



SYSTEMIC INFECTIONS

- ☑ HIV/AIDS
- Infective endocarditis
- Staphylococcal Toxic Shock Syndrome
- Staphylococcal Scalded Skin Syndrome
- Necrotizing Fasciitis
- Meningococcemia
- Rocky Mountain Spotted Fever
- Scarlet Fever
- Vibrio Infection
- Lyme Disease
- Acute Rheumatic Fever

INFECTIVE ENDOCARDITIS

- Skin lesions are caused by vascular or immunologic phenomenon
- Osler nodes
  - Tender purpuric nodules on the finger pads and toes
- Janeway lesions
  - Non-tender purpuric macules of the palms and soles.
- Splinter hemorrhages
  - Linear red streaks under the nail
- Treat underlying infection

Image source: VisualDx (visualdx.com)

Osler Nodes

Image source: VisualDx (visualdx.com)

Splinter Hemmorhages

Source: Kevin J. Knoop, Lawrence B. Stack, Alan B. Storrow, R. Jason Thurman: The Atlas of Emergency Medicine, 5e

Janeway Lesions

STAPHYLOCOCCAL TOXIC SHOCK & SCALDED SKIN SYNDROMES

Bacterial toxin-mediated skin disorders. Must be differentiated from Toxic Epidermal Necrolysis (TEN) and Steven Johnson's Syndrome (SJS)

Toxic Shock

- Systemic reaction to s. aureus toxins
- Acute onset diffuse cutaneous erythema, "strawberry tongue“, conjunctival injection, high fever, hypotension, negative Nikolsky, swelling of the hands and feet, followed by blistering and desquamation of the palms and soles after 1-3 weeks

Scalded Skin

- Skin reaction to toxin
- Bulla formation with large areas of desquamation, tenderness, erythema, exfoliation of skin, fever. Positive Nikolsky sign
- No mucous membrane involvement


Images source: VisualDx (visualdx.com)

NECROTIZING FASCIITIS

- Associated with group A *Streptococcus*, mixed aerobic–anaerobic bacteria, MRSA or can occur as a result of gas gangrene caused by *Clostridium perfringens*
- Can initially present as pain/fever prior to swelling and edema, with progression to hemorrhagic bullae characterized by large areas of sloughing that rupture to form rapidly enlarging areas of gangrene that extend down to the fascia causing brown-gray appearance.
- Urgent surgical exploration and debridement for this life-threatening infection.

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

MENINGOCOCCEMIA

- Acute meningococcemia can occur either in epidemics or in isolated cases
- Causative organism, Neisseria meningitidis, is usually identified in cerebrospinal fluid but can also be identified by smear or cultures of skin lesions or by blood cultures.
- Fever, headache, petechiae, ecchymoses, palpable purpura
- If untreated, develop DIC, with extensive hemorrhage, hypotension, and death.

Image source: VisualDx (visualdx.com)

Source: Gregory A. Schmidt, John P. Kress, Ivor S. Douglas: *Hall, Schmidt and Wood's Principles of Critical Care, 5th Edition*

ROCKY MOUNTAIN SPOTTED FEVER (RMSF)

- Tick-borne illness caused by *Rickettsia rickettsii*
- Characterized by the sudden onset of fevers, chills, and headache.
- Incubation period is 2-14 days after tick bite. Characteristic erythematous rash develops on the wrists and ankles and becomes purpuric, then spreads centrally to involve the extremities, trunk, and face. Palm and sole involvement is characteristic
- Treat empirically with Doxycycline if RMSF is suspected – treatment delay is associated with severe disease and increased mortality

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

SCARLET FEVER

- Scarlet fever begins with pharyngitis caused by group A β-hemolytic streptococci
- A generalized rash develops 2 to 3 days after onset of the pharyngitis
  - pinpoint erythematous papules that may be easier to palpate than to see
- Other characteristic lesions include strawberry tongue and linear petechial macules occurring in body folds (Pastia lines)
- As the rash fades, desquamation of hands and feet appears
- Treatment with penicillin results in rapid resolution of all symptoms

Source: R.P. Usatine, M.A. Smith, E.J. Mayeaux, Jr., H.S. Chumley
The Color Atlas and Synopsis of Family Medicine, Third Edition

Image source: DermNet (dermnetnz.org)

Image source: VisualDx (visualdx.com)

VIBRIO INFECTION

- *Vibrio vulnificus* infection arises from oral exposure or percutaneous exposure
- Symptoms: gastroenteritis, sepsis (with or without skin and soft-tissue infection), and primary wound infection.
- Skin manifestations: severe cellulitis with lymphangitis and necrotizing fasciitis. Patients can develop hemorrhagic bullae, with leukopenia and DIC.
- In chronic liver disease or immunocompromised, infection can occur after eating undercooked or raw seafood (i.e. oysters)
- Prompt treatment with antibiotics; management of complications may require intensive supportive care.

Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. Mcmichael, J.S. Orringer: Fitzpatrick's Dermatology, Ninth Edition

Vibrio vulnificus infection - hemorrhagic plaque and bullae in patient with diabetes and cirrhosis

LYME DISEASE

- Most common tick-borne disease caused by the spirochete *Borrelia burgdorferi* in the US. Transmitted primarily by the tick *Ixodes scapularis*
- Three stages: early, disseminated and late
- Characteristic skin lesion, erythema migrans, begins as an erythematous macule or papule at the site of the tick bite.
  - Over time - erythematous lesion expands to form a red ring, often with central clearing.
  - If left untreated, lesions last weeks or months. Hematogenous dissemination of spirochetes occurs after several weeks, resulting in multiple annular patches of erythema chronicum migrans
- Early localized disease - patients with erythema migrans and exposure history, treat without serologic confirmation

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

Disseminated disease with multiple confluent plaques

LET'S SOLVE THIS

Question 7: View and answer question in Yuja Video

RHEUMATIC FEVER

- Autoimmune reaction to infection with group A *Streptococcus*.
- Skin manifestations are relatively rare clinical features of rheumatic fever and include erythema marginatum and subcutaneous nodules
- Erythema marginatum: pink macules which central clearing; serpiginous, spreading edge. Usually occurs on trunk and limbs
- Subcutaneous nodules: non-tender, mobile nodules overlying bony prominences, approximately 1 cm in diameter; commonly occur on the extensor surfaces of elbows, hands, feet (sometimes vertebrae)

Image source: VisualDx (visualdx.com)

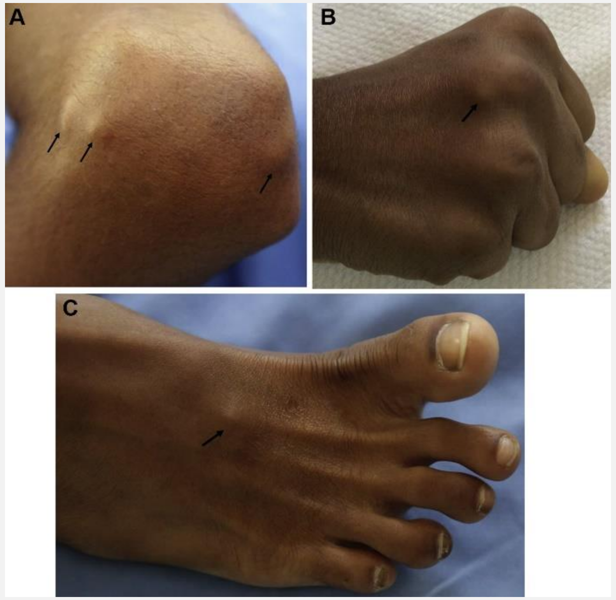
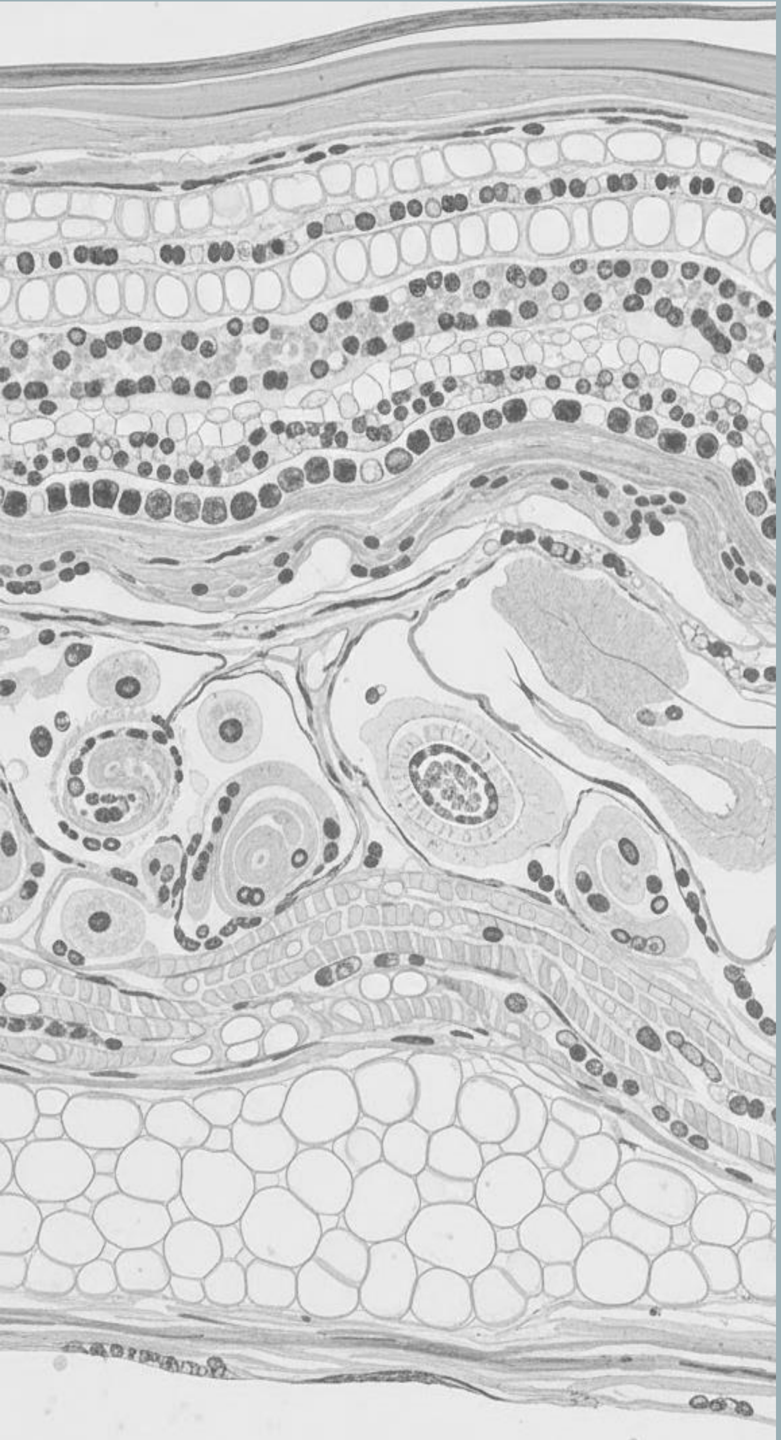


Image source: The Journal of Pediatrics, Volume 213, 242 - 242.e1



NEUROLOGIC

Neurofibromatosis

Tuberos Sclerosis

Basal Cell Nevus Syndrome

Incontientia Pigmenti

LET'S SOLVE THIS

Question 8: View and answer question in Yuja Video

NEUROFIBROMATOSIS TYPE I

- Neurofibromatosis is a group of 3 disorders (NF1, NF2, SWN) that present with tumors of central and peripheral nerve tissues
- NF1 is the most common autosomal genetic disorder
- Skin lesions for NF1:
  - Cutaneous neurofibromas, which are soft, skin-colored nodules that are often pedunculated
  - Café au lait macules are flat, evenly pigmented patches up to several centimeters in diameter.
  - Axillary freckling
- Plexiform neurofibromas are larger, deeper tumors that are associated with hypertrophy of bony and soft tissues – can become malignant peripheral nerve sheath tumors
- Lisch nodules—pigmented iris hamartomas—are also found in most patients with neurofibromatosis.

Axillary Freckling

Lisch nodules

Images source: VisualDx (visualdx.com)

TUBEROUS SCLEROSIS

- Tuberous Sclerosis Complex is a genetic disease caused by mutations *TSC1* or *TSC2* tumor suppressor genes which affect multiple organs including brain, eyes, kidney, heart, lungs, liver and skin.
- Skin lesions are present in most affected patients and include:
  - Hypomelanotic macules – ash leaf macules
  - “Confetti” lesions – smaller hypopigmented lesions
  - Facial angiofibromas – skin-colored papules on face
  - Shagreen patches – skin colored plaque consisting of thick dermal connective tissue
  - Periungual and subungual fibromas – skin-colored nodules around the fingers and toenails

Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. Mcmichael, J.S. Orringer: Fitzpatrick's Dermatology, Ninth Edition Copyright © McGraw-Hill Education. All rights reserved.

Image source: VisualDx (visualdx.com)

Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. Mcmichael, J.S. Orringer: Fitzpatrick's Dermatology, Ninth Edition Copyright © McGraw-Hill Education. All rights reserved.

Image source: VisualDx (visualdx.com)

BASAL CELL NEVUS SYNDROME

- The basal cell nevus syndrome is a rare, autosomal dominant disorder attributed to mutational inactivation of the *PTCH* gene **which increases risk**
- Skin findings:
  - Development of multiple basal cell carcinomas - pigmented or nonpigmented
  - Palmar pits (can also develop on soles)
  - Benign cysts in jaw
- Other findings include frontal bossing, broad nasal root, hypertelorism
- Can affect several systems and increased risk for developing medulloblastoma as well as cardiac or ovarian fibromas

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)


Source: Arturo P. Saavedra, Ellen K. Roh, Anar Mikailov
Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology, 9e

INCONTINENTIA PIGMENTI

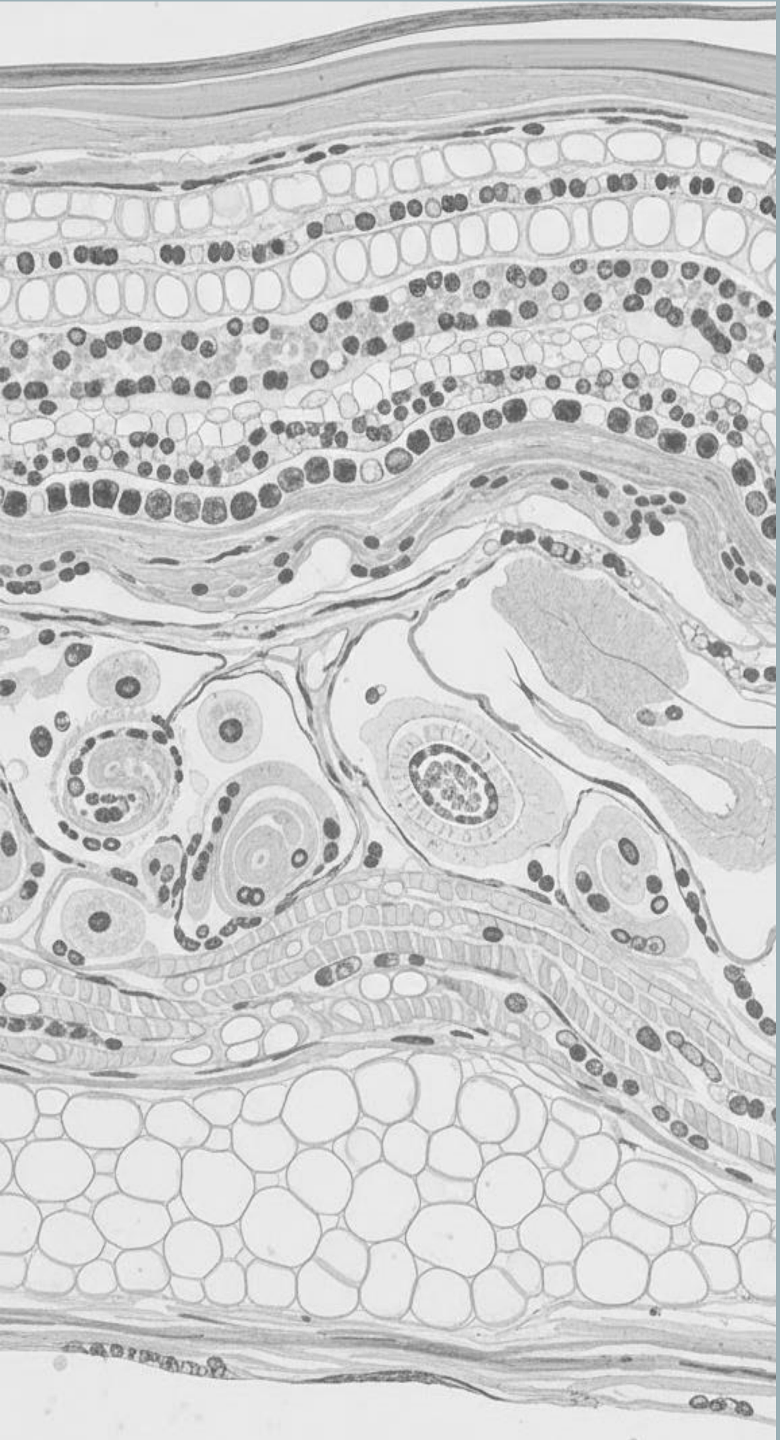
- Incontinentia pigmenti is an inherited syndrome that affects the skin and nervous system.
- Mutations in the *NEMO* gene, an essential component of the nuclear factor–kb signaling cascade, account for 85% of cases.
- The inheritance pattern is X-linked dominant and is lethal in males
- Disease progresses in 4 stages:
  - Vesicular
  - Verrucous – with linear plaques in extremities
  - Hyperpigmentation – linear pigmented whorls
  - Hypopigmentation/atrophy – associated with alopecia
- Associated with neurologic symptoms, abnormalities in skin, hair, nails, dental anomalies

Image source: VisualDx (visualdx.com)

Conical teeth

Image source: DermNet (dermnetnz.org)

Vesicular Stage



RENAL DISEASES

Fabry Disease

Polyarteritis Nodosa

Calciphylaxis

FABRY DISEASE

- X-linked lysosomal storage disease caused by a deficiency in activity of a-galactosidase A, resulting in deposition of glycosphingolipids in tissues
- Symptoms include hypohidrosis, telangiectasia, angiokeratomas as well as paresthesia
- **Angiokeratomas**
  - Dark red or purple macules or papules that resemble cherry hemangiomas
  - Most commonly found in the periumbilical area but can also occur on the palms, soles, trunk, extremities, and mucous membranes
  - Develop in childhood and increase over time as glycosphingolipids accumulate in various organs

Image source: VisualDx (visualdx.com)

LET'S SOLVE THIS

Question 9: View and answer question in Yuja Video

POLYARTERITIS NODOSA

- Necrotizing inflammation of small and medium sized muscular arteries, commonly associated with Hepatitis B
- Aneurysms form in many arteries, including those leading to the kidneys and subcutaneous tissue.
- Skin manifestations include: Livedo reticularis, cutaneous ulcers
- An isolated cutaneous form presents with painful nodules of the lower extremities; systemic signs may be present but less common. Often starts in childhood.

Image source: VisualDx (visualdx.com)

Image source: VisualDx (visualdx.com)

CALCIPHYLAXIS

- Calciphylaxis (calcific uremic arteriolopathy), is a life-threatening condition in end-stage renal disease (ESRD) patients. Warfarin use is an additional risk factor
- Calcifying vasculopathy which affects blood vessels in the dermis and subcutaneous tissues.
- Begins with painful purpuric plaques and nodules that may be reticulated, which progress to indurated plaques, bulla formation and may ulcerate, becoming necrotic
- Ulcers with black eschars can be very painful
- Prolonged clinical course with poor prognosis

Image source: VisualDx (visualdx.com)

Retiform, hemorrhagic crust, surrounding dusky patches

Source: Carol Soutor, Maria K. Hordinsky:
Clinical Dermatology: Diagnosis and Management of Common I

Deep ulcer with black eschars

REFERENCES

Pearson DR. Skin Signs of Systemic Disease. In: Soutor C, Hordinsky MK. eds. *Clinical Dermatology: Diagnosis and Management of Common Disorders, 2e*. McGraw-Hill Education; 2022.

Cheeley JT, Lawley TJ, Stoff BK. Atlas of Skin Manifestations of Internal Disease. In: Longo D, Fauci A, Kasper D, Hauser S, Jameson J, Loscalzo J, Holland S, Langford C. eds. *Harrison's Principles of Internal Medicine, 22nd Edition*. McGraw Hill; 2026.

Varga J. Systemic Sclerosis (Scleroderma) and Related Disorders. In: Longo D, Fauci A, Kasper D, Hauser S, Jameson J, Loscalzo J, Holland S, Langford C. eds. *Harrison's Principles of Internal Medicine, 22nd Edition*. McGraw Hill; 2026.

Hardin J. Infective Endocarditis. In: Knoop KJ, Stack LB, Storrow AB, Thurman R. eds. *The Atlas of Emergency Medicine, 5e*. McGraw-Hill; 2021.

Hasegawa A, Kimura H, Abe R. Dermatologic Conditions. In: Schmidt GA, Kress JP, Douglas IS. eds. *Hall, Schmidt and Wood's Principles of Critical Care, 5th Edition*. McGraw Hill; 2023.

Norton SA, Cardis MA. Miscellaneous Bacterial Infections with Cutaneous Manifestations. In: Kang S, Amagai M, Bruckner AL, Enk AH, Margolis DJ, McMichael AJ, Orringer JS. eds. *Fitzpatrick's Dermatology, 9e*. McGraw-Hill Education; 2019

Ngan, Vanessa, Incontentia Pigmenti; https://dermnetnz.org/topics/incontinentia-pigmenti. Accessed Nov 15, 2025

Kurta AO, Glaser D. Hyperhidrosis and Anhidrosis. In: Kang S, Amagai M, Bruckner AL, Enk AH, Margolis DJ, McMichael AJ, Orringer JS. eds. *Fitzpatrick's Dermatology, 9e*. McGraw-Hill Education; 2019.

Venkatesh KP, Raza MM, Nickel G, Wang S, Kvedar JC. Deep learning models across the range of skin disease. NPJ Digit Med. 2024 Feb 10;7(1):32. doi: 10.1038/s41746-024-01033-8. PMID: 38341516; PMCID: PMC10858968.

Hasanzadeh, F., Josephson, C.B., Waters, G. *et al*. Bias recognition and mitigation strategies in artificial intelligence healthcare applications. *npj Digit. Med.* **8**, 154 (2025). https://doi.org/10.1038/s41746-025-01503-7

Kololgi SP, Lahari CS. Harnessing the Power of Artificial Intelligence in Dermatology: A Comprehensive Commentary. Indian J Dermatol. 2023 Nov-Dec;68(6):678-681. doi: 10.4103/ijd.ijd_581_23. Epub 2024 Jan 9. PMID: 38371574; PMCID: PMC10868991.

AI Demystified: Introduction to large language models. Updated December 13, 2024. Accessed November 22, 2025. https://uit.stanford.edu/service/techtraining/ai-demystified/llm

Brief Definitions of Key Terms in AI. Accessed November 22, 2025. https://hai.stanford.edu/policy/brief-definitions-of-key-terms-in-ai

Anshelevich E, Rashighi M, Tan B,Burgin S. Subacute Cutaneous Lupus Erythematosus. VisualDx. Last Updated:09/22/2025. Accessed November 22, 2025. https://www.visualdx.com/visualdx/diagnosis/?moduleId=101&diagnosisId=51885&imgSet=1#synopsis

Google Gemini 3.0. (2025). Accessed November 22, 2025. https://gemini.google.com

QUESTIONS?

Feel free to email ybains@touro.edu